

ShadowMaker: Sketch-Based Creation Tool for Digital Shadow Puppetry

Zhihao Yao
Tsinghua University
Beijing, China
yaozh_h@outlook.com

Shiqing Lyu
Tsinghua University
Beijing, China
lvsq22@mails.tsinghua.edu.cn

Yao Lu
Tsinghua University
Beijing, China
linaluyao@gmail.com

Qirui Sun
Tsinghua University
Beijing, China
sunqr97@gmail.com

Hanxuan Li
Tsinghua University
Beijing, China
lihanxua22@mails.tsinghua.edu.cn

Xuezhu Wang
Tsinghua University
Beijing, China
15626222995@163.com

Guanhong Liu
Tongji University
Shanghai, China
liugh23@tongji.edu.cn

Haipeng Mi
Tsinghua University
Beijing, China
mhp@tsinghua.edu.cn

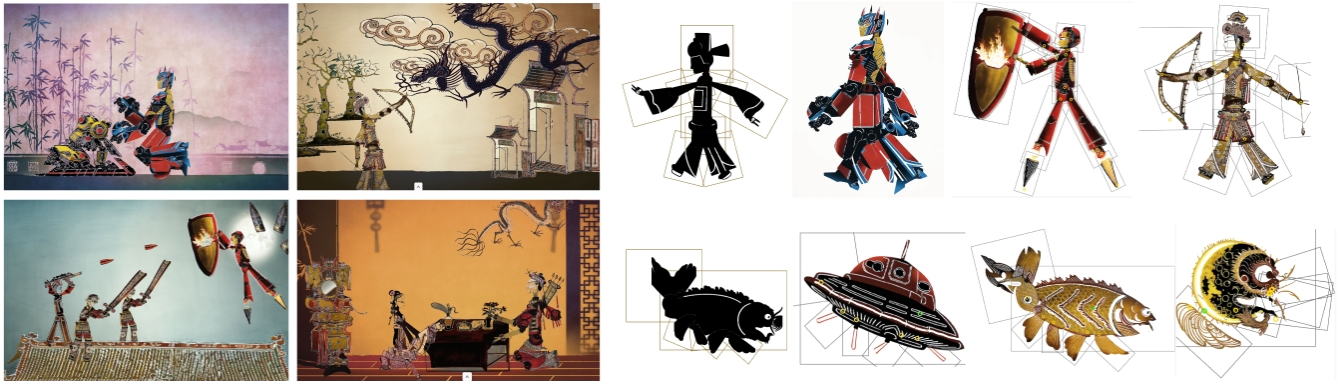


Figure 1: Artworks created with Shadowmaker(left). A variety of thematic elements created using sketch templates (right).

ABSTRACT

Shadow puppetry is a cultural heritage with thousands of years of history, and its digital form has become a popular style in modern artistic creation. Creating high-quality shadow puppet works is often challenging. We present "ShadowMaker," an intelligent collaborative tool, where designers can co-create dynamic digital shadow puppets together with AI through prompts and sketching. ShadowMaker introduces a novel non-linear workflow, allowing users to freely switch between stages like sketching, layout, and animating. The user study with 18 participants demonstrated that ShadowMaker is easy to use, offers a high degree of creative freedom, and is capable of inspiring users' creativity and inspiration.

Permission to make digital or hard copies of part or all of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).
CHI EA '24, May 11–16, 2024, Honolulu, HI, USA
© 2024 Copyright held by the owner/author(s).
ACM ISBN 979-8-4007-0331-7/24/05
<https://doi.org/10.1145/3613905.3648657>

CCS CONCEPTS

• Human-centered computing → User interface toolkits.

KEYWORDS

digital shadow puppetry, authoring, sketching, Creative Support Tools

ACM Reference Format:

Zhihao Yao, Shiqing Lyu, Yao Lu, Qirui Sun, Hanxuan Li, Xuezhu Wang, Guanhong Liu, and Haipeng Mi. 2024. ShadowMaker: Sketch-Based Creation Tool for Digital Shadow Puppetry. In *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems (CHI EA '24)*, May 11–16, 2024, Honolulu, HI, USA. ACM, New York, NY, USA, 5 pages. <https://doi.org/10.1145/3613905.3648657>

1 INTRODUCTION

Shadow puppetry, a traditional art form originating in China, is recognized as a traditional craftsmanship in urgent need of protection on the Intangible Cultural Heritage list. With the passage of time and the evolution of media, shadow puppetry has transformed into a unique digital art style, finding extensive applications in fields

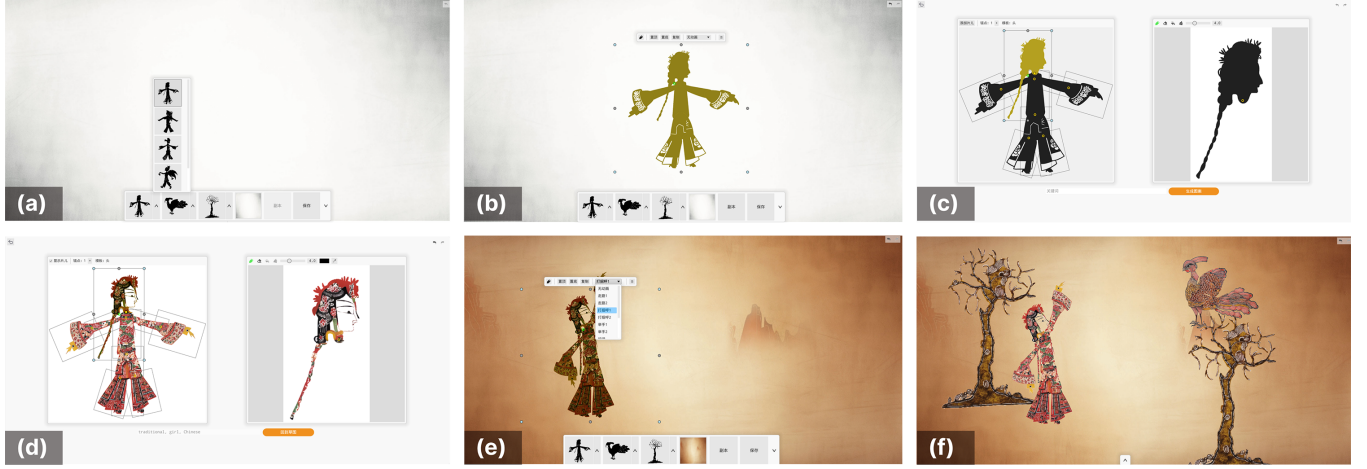


Figure 2: Walkthrough of creating the work 'Farewell' using Shadowmaker

like animation, film, and gaming[8, 9]. Creating scenes in shadow puppetry is often challenging: it involves multiple aspects such as characters, settings, and props, and the production process is complex, requiring steps like sketching, character design, animation, and overall layout.

In the digital shadow puppetry field, some studies have utilized template libraries[5, 7] to provide innovative tools for the generation of shadow puppet characters. Additionally, research focusing on using motion-sensing devices for motion capture and real-time control has also been conducted[3, 4]. While these studies have made progress in specific aspects of shadow puppetry, they have not yet culminated in a comprehensive creative tool. Lin et al. provided a more comprehensive creation process, using portrait photos to generate shadow puppet-like heads and employing animation templates for narrative creation[6]. However, this method is limited to only the head portion and has a singular theme, which greatly limits the freedom in creative expression and the diversity of narrative themes. To our knowledge, there are currently no tools available that support shadow puppetry creation across a broad range of themes, which we consider fundamental for freeform creation. On the other hand, AI-based tools often overlook the level of creator involvement, leading to the creator's intent not being well-represented in the final work.

In response to the studies and challenges previously mentioned, we introduce ShadowMaker, an AI-assisted creative system that integrates sketching and prompts. This tool enables users to easily bring their creative ideas to life and build complete, dynamic shadow puppet scenes. We invited a small group of users to create with our tool, resulting in a diverse range of artworks as shown in Figure 1. Our contributions can be summarized as follows:

- **Algorithm:** We present a novel dynamic shadow puppet generation framework, enabling the direct transformation of sketches into drivable shadow puppets.
- **System:** We developed "ShadowMaker," a web-based creative tool offering an innovative non-linear workflow. This allows users to flexibly and quickly switch between various functions, facilitating rapid iteration of ideas. The tool

includes a diverse set of features catering to different skill levels, enabling users to engage in highly creative work with a high degree of freedom.

- **Data:** We have curated a high-quality digital shadow puppet dataset and trained multiple AI models. By employing a stacked multi-model approach, we achieve superior visual results.

2 USER INTERFACE AND INTERACTION

2.1 Walkthrough

For easier understanding, we first demonstrate the complete process of creating a work titled "Farewell," as shown in Figure 2. The creator begins by selecting a shadow puppet template from the menu (Figure 2a), then uses tools to position and scale it (Figure 2b). After clicking the edit button to access the puppet's editing interface (Figure 2c), the creator modifies the sketch of a selected component. A prompt is inputted at the bottom, and upon clicking 'generate', the result is obtained (Figure 2d). Then, returning to the scene interface, the creator selects a waving animation and a preset background (Figure 2e). Following similar steps, two trees are added to complete the scene arrangement (Figure 2f). It is worth noting that some users prefer to first arrange the layout and animation before delving into individual object editing. The workflow is flexible and not strictly linear, allowing users to choose their preferred order of steps.

2.2 Scene Interface

The scene interface is the initial interface when users open the tool, divided into a canvas and a menu floating at the bottom (Figure 3 left).

Within the canvas, upon selecting a shadow puppet, a toolbar appears, offering transformation controls for position, rotation, and scaling, as well as options for layering adjustments (Figure 3a). Additionally, there is an animation bar that allows users to apply species and structure-specific animations to the puppet (Figure 3b).

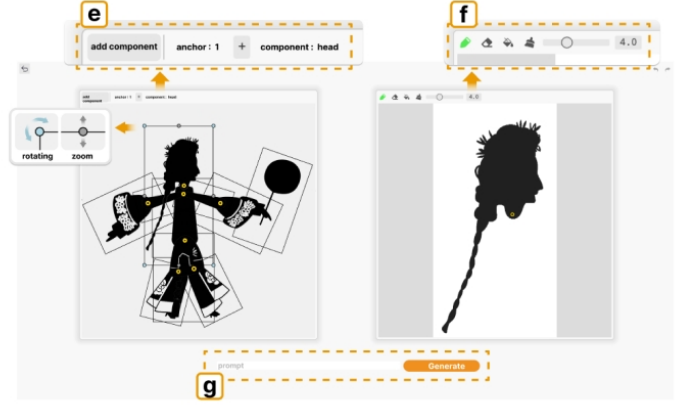


Figure 3: Scene Interface(left) and Editing Interface(right)

The Bottom Menu offers four common templates including humanoid, animal, props, and backgrounds(Figure 3c). These templates, designed by professional designers, are detailed in Figure 4. The Instance List(Figure 3d), accessible via the instance button, displays all shadow puppet objects in the current scene, allowing users to create duplicates that remain synchronized with the original, except for animations. This facilitates the creation of repetitive elements like vegetation. The Save and Load feature enables users to save the scene as a local JSON file for backup(Figure 3d), collaboration, and maintaining editability, including sketches, templates, and animations. Users can load existing projects by dragging them into the webpage.

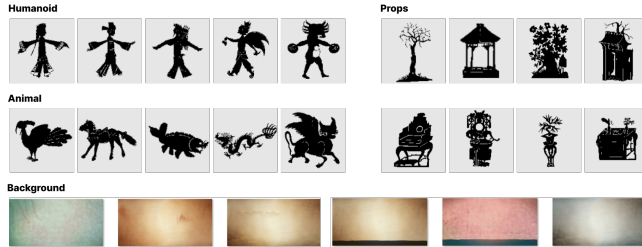


Figure 4: Templates library

2.3 Editing Interface

By double-clicking any shadow puppet in the scene interface or clicking the edit button, users can enter the editing interface for the corresponding shadow puppet(Figure 3 right). The editing interface consists of three parts: the component panel on the left, the local canvas on the right, and the text input box at the bottom, as shown in Figure 3. The component panel displays all components of the shadow puppet, allowing users to modify individual components and their connection relationships (Figure 3e). The local canvas updates according to the selected component, where users can sketch and modify the final generated result (Figure 3f). By entering their ideas in the input box, users can initiate the generation process (Figure 3g).

Each template comprises non-removable core components and auxiliary components that users can freely add or delete. Every component can be modified, with rotation and scaling achieved through control points on the square border. Users can freely add components to create their own personalized templates. The figure 5 illustrates the entire process of modifying a human template to have multiple arms.

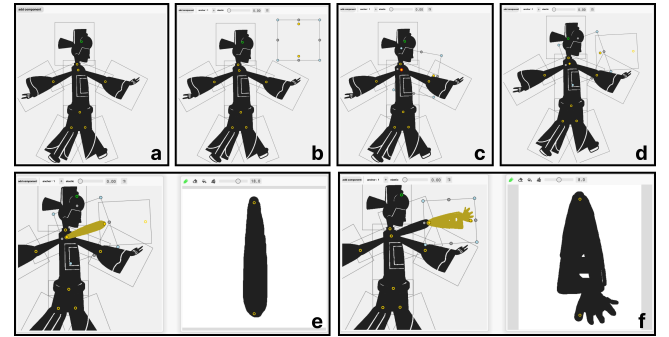


Figure 5: The whole process of adding an arm to the default character. All components are within the panel (a). By clicking the 'Add Component' button, the component will appear in the panel (b). Drag the new component onto an existing one to connect them (c). Repeat the process to obtain components for both the upper arm and the forearm (d). After selecting the upper arm component, draft on the local board on the right (e); perform the same operation for the forearm component (f).

3 IMPLEMENTATION

ShadowMaker is a web-based application that can run on various devices. Our technical framework mainly consists of three parts: generation algorithm, animation production and application, and the user interface and tools, as shown in Figure 6.

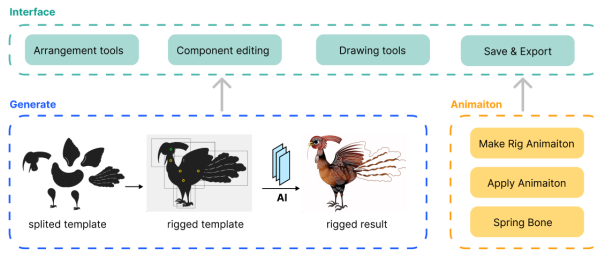


Figure 6: Implementation Framework

3.1 Shadow Puppet Generation

3.1.1 AI model. We utilized the stable diffusion large model combined with LoRA fine-tuning for two rounds of training. In the first training session, we collected and processed 200 images from open-source digital museums, aiming to encompass a wide variety of objects and characters, thereby enabling the AI to generate diverse themes. For the second training, we selected and re-drew over 20 high-quality shadow puppet images, focusing on refining and learning a shadow puppet style that aligns with modern aesthetics. Ultimately, we successfully constructed two LoRA models, each tailored for generating a broad range of themes and for high-quality style transfer, respectively.

3.1.2 Image Processing. The system first assembles individual components into a complete rigged template for AI generation, followed by segment-specific cropping to ensure uniform styling. Upon user command, it draws component silhouettes onto a single image, sends it to the Stable Diffusion server with preset parameters for text-to-image generation, and then performs image segmentation and matting to extract and refine each component's colored silhouette from the background.

3.2 Development of Interface

We developed the interface and tools using Rust, with a cross-platform GUI using egui and compiled it to WebAssembly for web browser compatibility. The tool utilizes SVG files to construct templates, where designers create shadow puppets, annotate anchor points and components, and export them as SVG. The tool then parses SVG markup into Rust code, reducing workload for material updates. Computational tasks are executed on the CPU, implementing algorithms for drawing, brushes, and erasers. For fill color and erasure, floodfill is used, and a color picker tool with a magnifying glass panel is provided for touchscreen devices.

3.3 Animation

We utilize Blender, an open-source 3D animation and modeling software to produce animation data. We import static SVG files to create skeletons, where each component corresponds to a bone and anchor points represent connections between bones. We store various action sequences as binary glTF format files (glb) within Blender, using constraints to restrict component rotations to the local coordinate system's z-axis. Animation Mapping applies these animations to core components, with auxiliary components following the core components' movement as child elements. Elastic

Motion adds swaying and elasticity to auxiliary components while tracking the core components' movements.

4 USER STUDY

We conducted a user experiment with 18 participants (9 males, 9 females) whose ages ranged from 18 to 35 years old. All of them were designers with an interest in shadow puppets. The purpose of this study was to assess the usability and creative support capabilities of ShadowMaker. Before the experiment began, participants were provided with instructional videos to familiarize themselves with the tool, ensuring they had the necessary skills for free creative experimentation. After completing their creations, participants were asked to fill out the System Usability Scale (SUS)[1] and Creative Support Index (CSI)[2] surveys. The SUS survey aimed to evaluate ShadowMaker's usability and learnability, while the CSI survey assessed ShadowMaker's ability to assist users in creative work.

Following the questionnaire completion, semi-structured interviews were conducted with the participants. Initially, participants were asked to introduce their creations and describe their creative inspirations. Subsequently, discussions revolved around pain points, freedom of expression, and immersion during the creative process. All participants successfully completed the creative experimentation task.

All participants completed their creations. Some artworks are shown in Figure 1. The overall SUS survey score was 75.52 (sd=12.51) out of 100, indicating that our tool performed well in terms of usability. The CSI survey yielded an overall score of 82.33 (sd=11.93) out of 100, signifying that our tool demonstrated outstanding creative support capabilities. Participants unanimously believed that our tool inspired their creative ideas and even led to the creation of things they had never imagined before.

5 CONCLUSION

In this study, we introduced "ShadowMaker", a sketch-based creation tool for shadow puppetry, allowing for high-freedom dynamic scenarios. We explored an automated rigging algorithm using pre-segmented templates, enabling the assembly of a static image onto a dynamic skeleton. Available to users as a web-based creative tool, it allows them to modify templates, freely sketch while combining with prompts to generate desired shadow figures, and animate them with a single click. We will enhance user research to further discuss the human-machine collaborative model within the context of traditional culture.

REFERENCES

- [1] Aaron Bangor, Philip T Kortum, and James T Miller. 2008. An empirical evaluation of the system usability scale. *Intl. Journal of Human-Computer Interaction* 24, 6 (2008), 574–594.
- [2] Erin Cherry and Celine Latulipe. 2014. Quantifying the Creativity Support of Digital Tools through the Creativity Support Index. *ACM Trans. Comput.-Hum. Interact.* 21, 4, Article 21 (jun 2014), 25 pages. <https://doi.org/10.1145/2617588>
- [3] Tang Chuanqian, Liu Zhiqiang, Su Yijun, and Liu Xiaojing. 2022. Research on Real-Time Motion Matching of Shadow Play Based on Kinect. *Journal of System Simulation* 34, 3 (2022), 633.
- [4] Teng Li and Weiqun Cao. 2021. Research on a method of creating digital shadow puppets based on parameterized templates. *Multimedia Tools and Applications* 80 (2021), 20403–20422.
- [5] Hui Liang, Jian Chang, Meili Wang, Can Chen, and Jian Jun Zhang. 2016. Semantic framework for interactive animation generation. In *Proceedings of the 15th*

ACM SIGGRAPH Conference on Virtual-Reality Continuum and Its Applications in Industry-Volume 1. 137–145.

- [6] Min Lin, Zhenzhen Hu, Si Liu, Meng Wang, Richang Hong, and Shuicheng Yan. 2013. eHeritage of shadow puppetry: creation and manipulation. In *Proceedings of the 21st ACM international conference on Multimedia*. 183–192.
- [7] Argyro Moraiti, Nektarios Moutoutzis, Marios Christoulakis, Andreas Pitsiladis, George Stylianakis, Yiannis Sifakis, Ioannis Maragoudakis, and Stavros Christodoulakis. 2016. Playful creation of digital stories with eShadow. In *2016 11th International Workshop on Semantic and Social Media Adaptation and Personalization (SMAP)*. IEEE, 139–144.
- [8] Zhang Qing. 2020. The Interactive Expression and Language Transformation in the Application of VR Technology to Chinese Museum Exhibitions—Take the Development and Application of VR Shadow-play Game "Tian Ji's Horse Racing" as an Example. *Chinese Museum*. 2 (2020).
- [9] XU Zi-han. 2015. Development of Chinese shadow play:a study from the perspective of art communication. *Journal of Southeast University(Philosophy and Social Science)*. 6 (2015), 127–134.